

Time Series

Forecasting Athlete Performance Score

Dataset: Athlete Performance Dataset | 1,000 rows | 16-Hour Window

Dataset Overview & Time Series Exploration

1,000

Observations

1 min

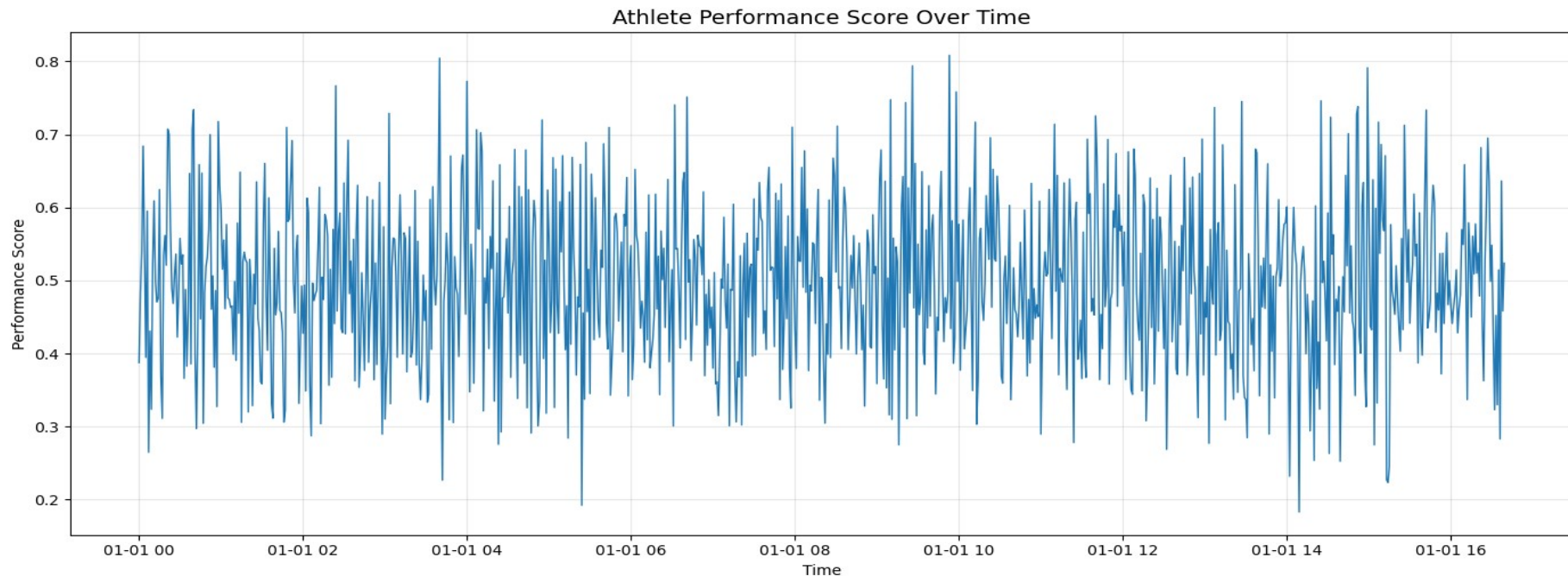
Interval

0

Missing Values

Stationary

ADF Test Result



Performance score oscillates between 0.18–0.81 with no clear trend or seasonality over the 16-hour window.

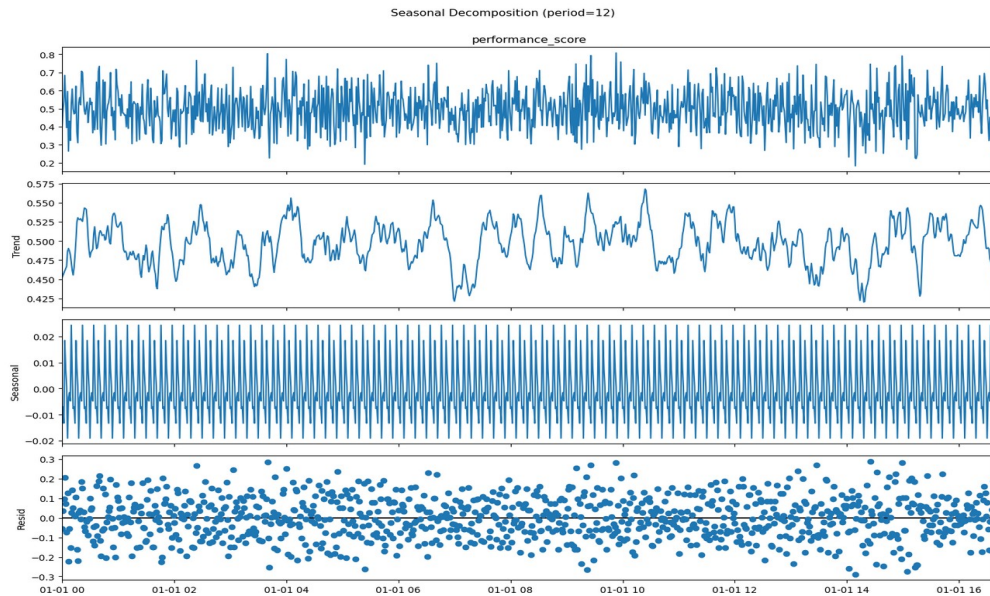
Stationarity Test & Seasonality Detection

ADF Stationarity Test

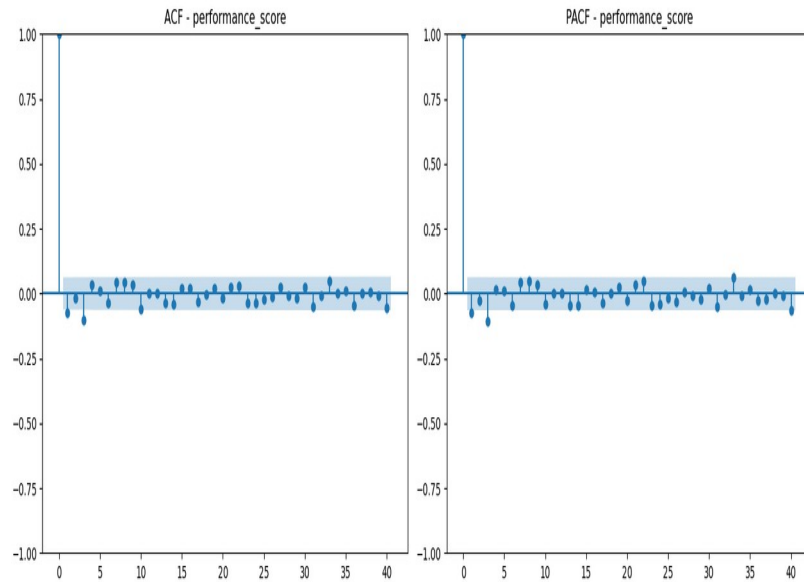
ADF Statistic	-21.05
p-value	≈ 0.0000
Result	Stationary ✓

Seasonality Detection

- Seasonal decomposition: no meaningful seasonal component
- ACF/PACF: no significant autocorrelation beyond lag 0
- Conclusion: series behaves like white noise



Seasonal Decomposition (period=12)



ACF & PACF — no significant autocorrelation

Forecasting Methods & Pipeline

Chronological Split: Training 80% (800 obs) | Test 20% (200 obs) | *No shuffle — temporal order preserved*

Classical Method

Method: Holt-Winters Exponential Smoothing

Configuration: Additive trend, no seasonality

Training set: 800 observations

Forecast horizon: 200 test points

Why no seasonal?: No meaningful pattern detected

ML Method

Method: Random Forest Regressor

Features: lag_1, lag_2, lag_3, lag_7, rolling mean/std

Preprocessing: StandardScaler in Pipeline

Training set: 794 rows (after lag creation)

Cross-validation: TimeSeriesSplit (no leakage)

★ *No shuffle on time series data — always split chronologically*

Hyperparameter Tuning — Random Forest

GridSearchCV Setup

Method	GridSearchCV with TimeSeriesSplit
Model	Random Forest Regressor
CV Folds	TimeSeriesSplit (training only)
Combinations tested	108

Parameter Grid

- `n_estimators`: [100, 200, 300]
- `max_depth`: [None, 5, 10]
- `min_samples_split`: [2, 5, 10]
- `min_samples_leaf`: [1, 2]

Best Parameters Found

<code>n_estimators</code>	300
<code>max_depth</code>	None (unlimited)
<code>min_samples_split</code>	2
<code>min_samples_leaf</code>	1

Results

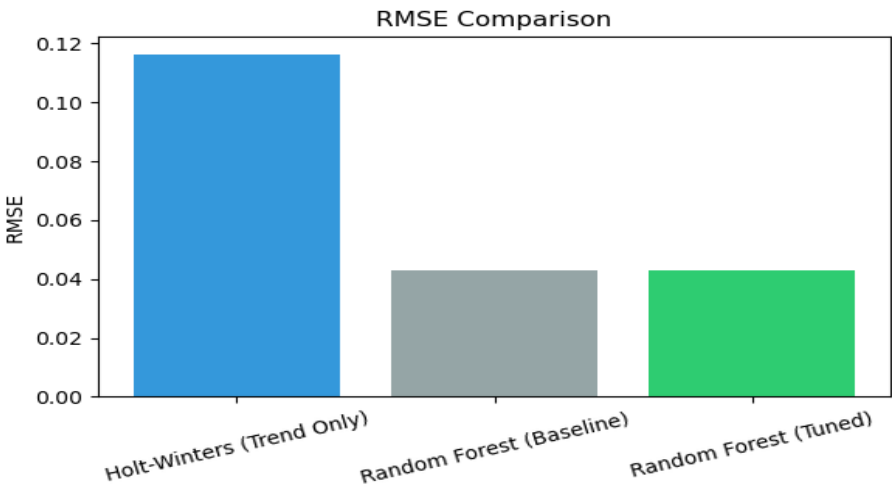
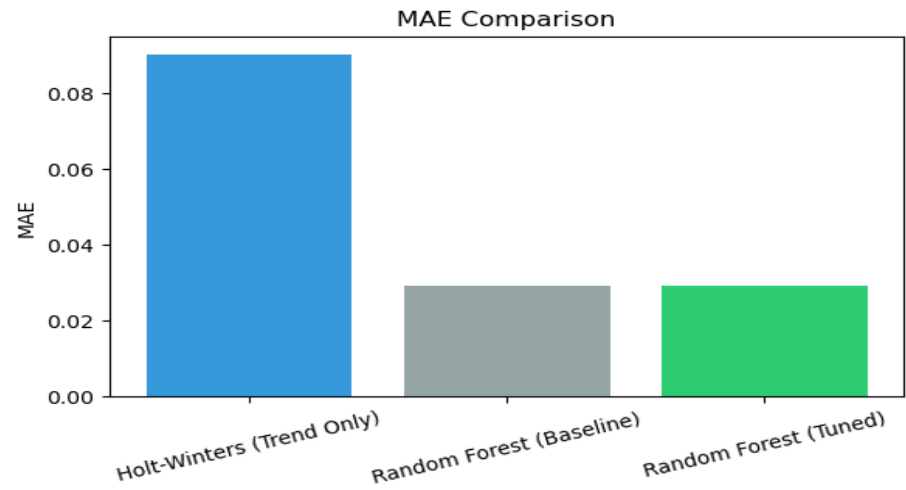
Default RF → MAE: 0.0294 RMSE: 0.0428
Tuned RF → MAE: 0.0294 RMSE: 0.0428

Default settings were already optimal — tuning confirmed the configuration.

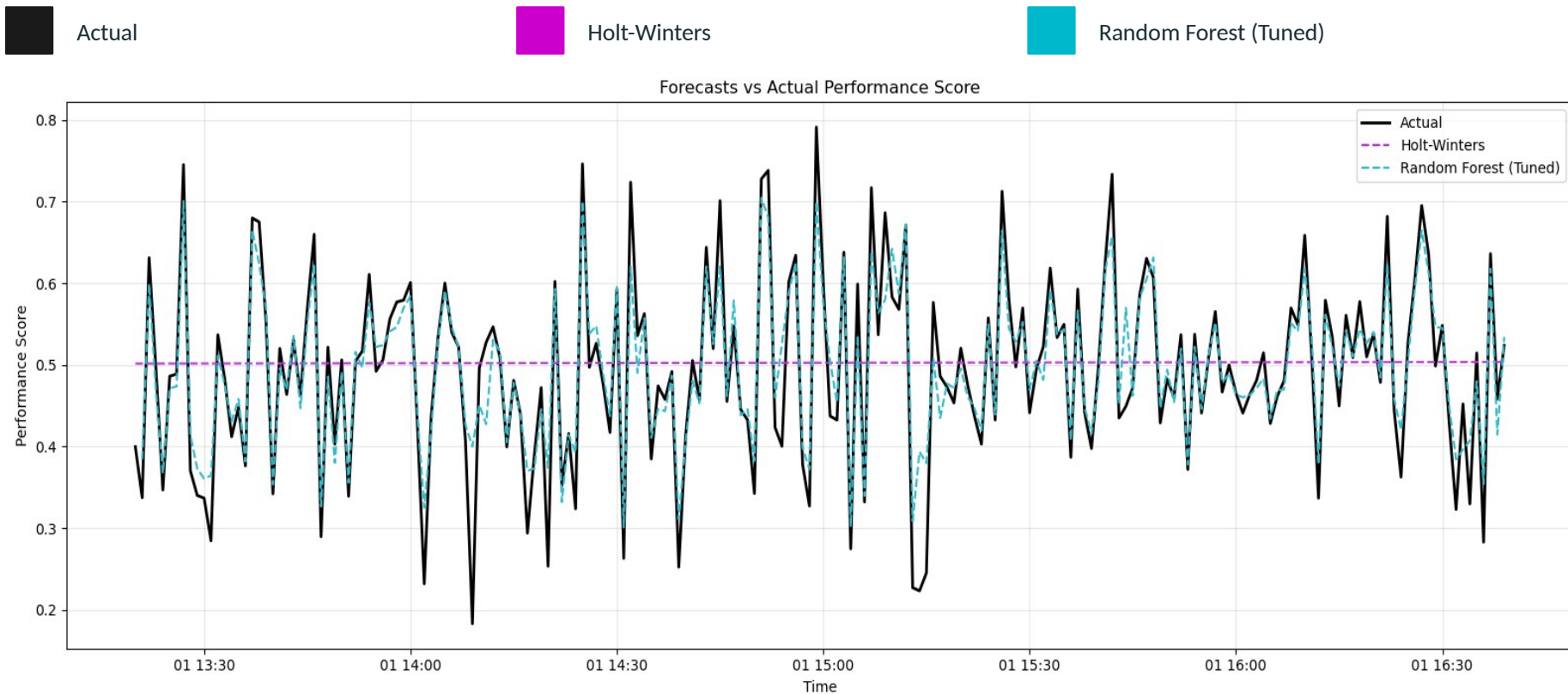
Model Performance Results

Method	MAE	RMSE
Holt-Winters (Trend Only)	0.0904	0.1164
Random Forest (Baseline)	0.0294	0.0428
Random Forest (Tuned) ★	0.0294	0.0428

★ Best Model



Forecasts vs. Actual Performance Score



Random Forest tracks the actual values closely; Holt-Winters reverts to a flat mean (~0.50).

Method Comparison & Practical Trade-Offs

Holt-Winters

Pros:

No feature engineering required; easy to interpret; fast to train

Cons:

Cannot capture short-term lag patterns; flat forecast when no trend

MAE:

0.0904

RMSE:

0.1164

Best for:

Data with clear trend/seasonality; interpretable dashboards

Random Forest (Tuned)

Pros:

Captures lag patterns; robust to noise; best accuracy

Cons:

Requires lag feature engineering; less interpretable

MAE:

0.0294

RMSE:

0.0428

Best for:

Real-time athlete monitoring; complex short-term patterns

Conclusion

3 Models Tested

Holt-Winters
Random Forest (Baseline)
Random Forest (Tuned)

Best Model — Random Forest (Tuned)

MAE: 0.0294 | RMSE: 0.0428
Captures short-term lag patterns effectively

Tuning Impact

GridSearchCV tested 108 combinations
300 trees, no depth limit — same as default
Confirmed default RF was already optimal

Key Finding

The series is white noise with no seasonality
ML with lag features outperforms classical methods
Random Forest recommended for real-time use